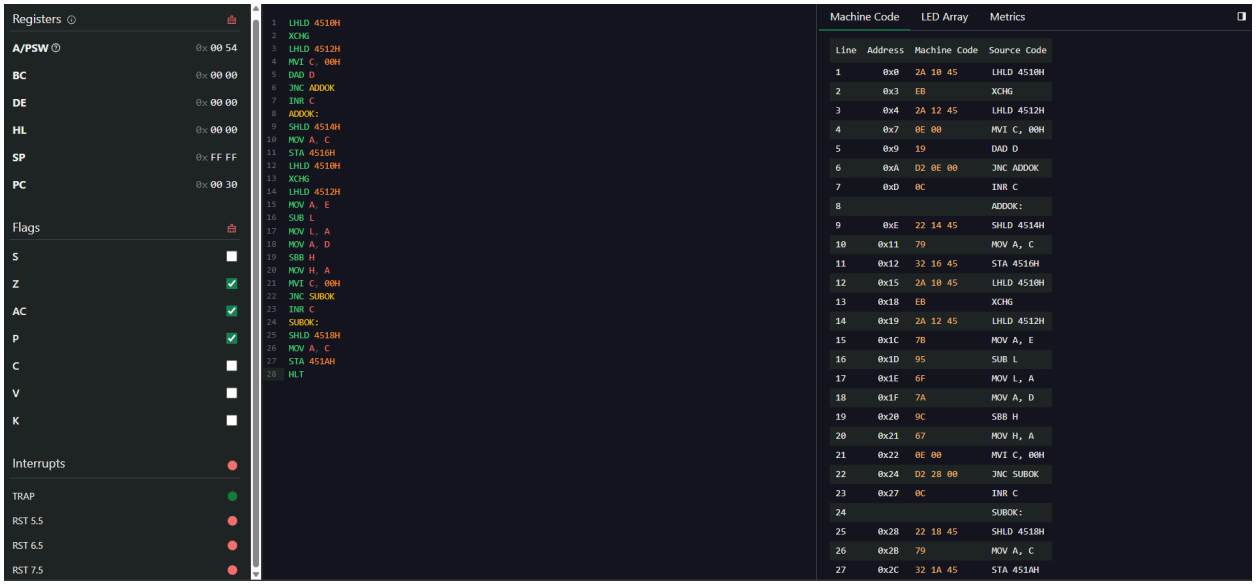
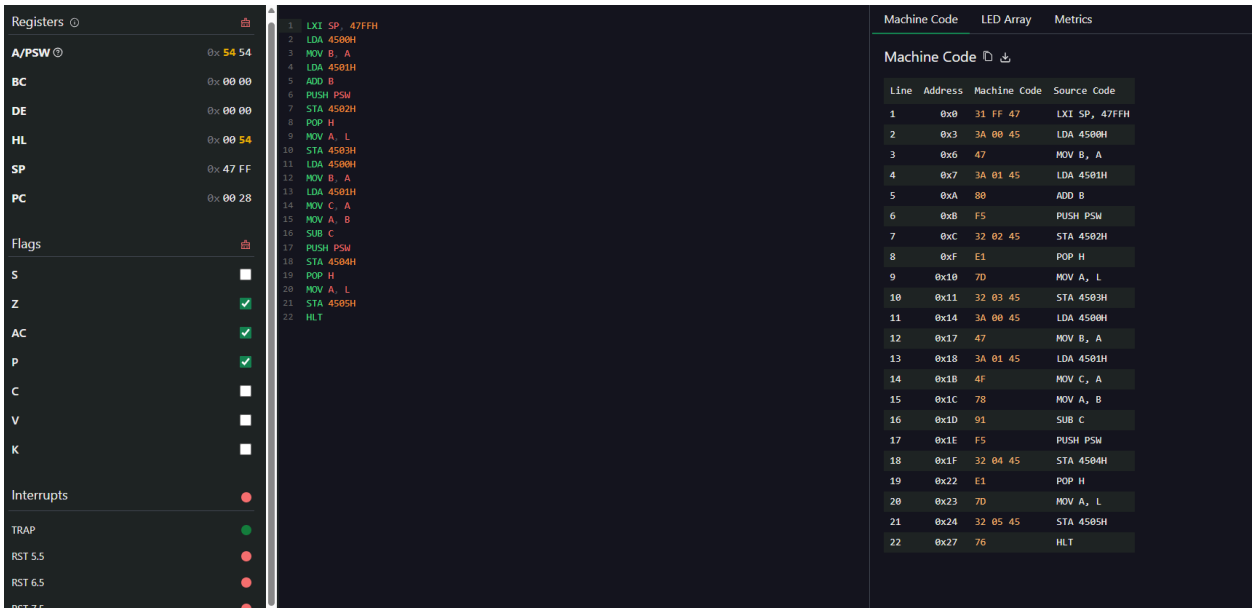
1. Develop and execute an 8-bit Microprocessor program to add and subtract two 8-bit numbers A = 5AH, B = 27H and two 16-bit numbers X = 1234H, Y = 4321H stored in memory. Store all results in consecutive memory locations, record the carry/borrow for each operation and validate the results by manual hexadecimal calculation. Repeat for A = E8H, B = 9CH, X = F234H, Y = 89ABH.

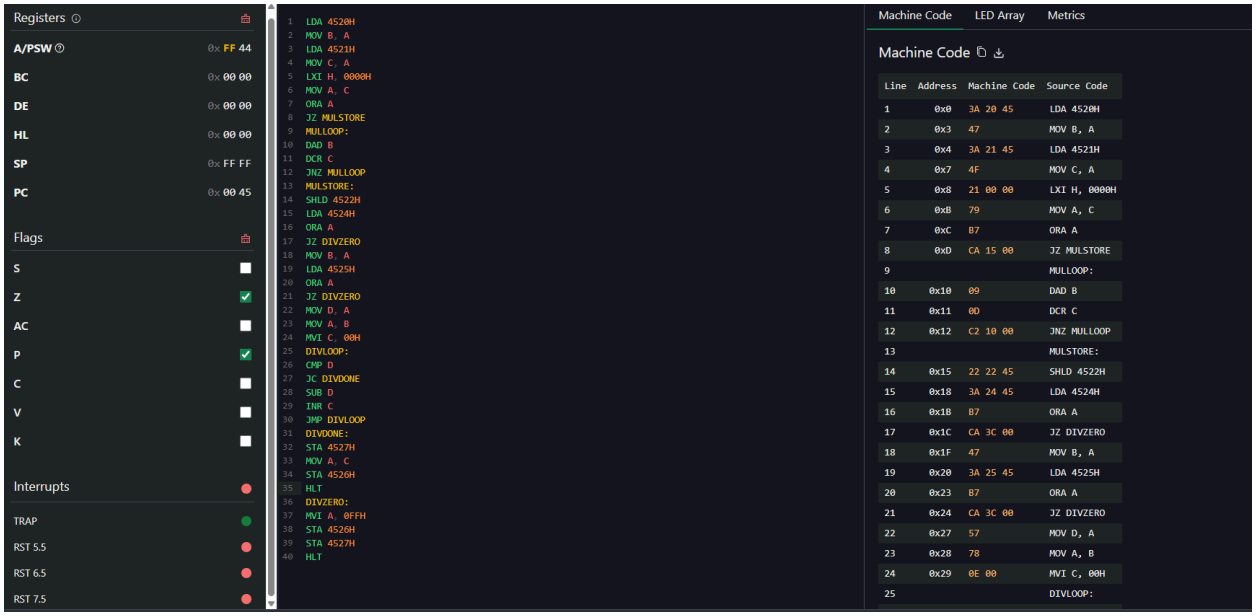


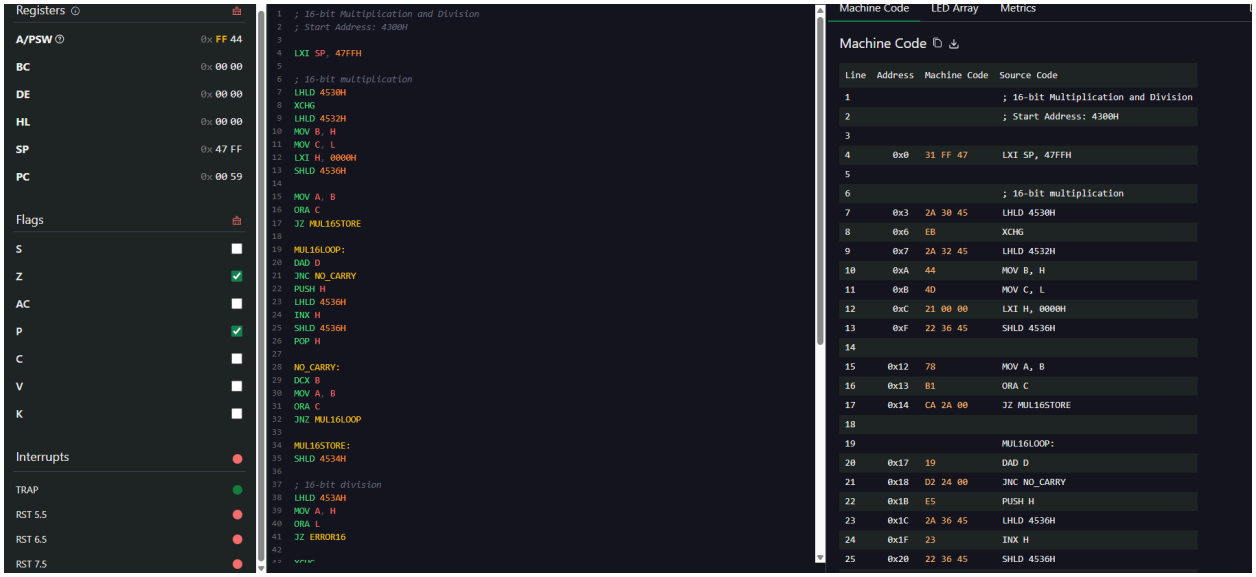


2. For A = E8H, B = 9CH record S, Z, AC, P and CY after addition and subtraction using PUSH PSW, find signed overflow manually, identify the cases where CY and overflow differ, and justify each flag using the signed and unsigned interpretation of the operands.



3. Develop an 8-bit Microprocessor program to perform 8-bit multiplication and division (A = 14H, B = 06H) and 16-bit multiplication and division (X = 1357H, Y = 0025H) using repeated addition and repeated subtraction. Store the product, quotient and remainder separately and compare the number of loop iterations in each case. Repeat for A = F2H, B = 0AH, X = 9A23H, Y = 0011H.


Registers
A/PSW 0x FF 44
BC 0x 00 00
DE 0x 00 00
HL 0x 00 00
SP 0x FF FF
PC 0x 00 45
Flags
S
Z
AC
P
C
V
K
Interrupts
TRAP
RST 5.5
RST 6.5
RST 7.5
LDA 4520H
MOV B, A
LDA 4521H
MOV C, A
LXI H, 0000H
MOV A, C
ORA A
JZ MULSTORE
MULLOOP:
DAD B
DCR C
JNZ MULLOOP
MULSTORE:
SHLD 4522H
LDA 4524H
ORA A
JZ DIVZERO
MOV B, A
LDA 4525H
ORA A
JZ DIVZERO
MOV D, A
MOV A, B
MVI C, 00H
DIVLOOP:
CMP D
JC DIVDONE
SUB D
INR C
JMP DIVLOOP
DIVDONE:
STA 4527H
MOV A, C
STA 4526H
HLT
DIVZERO:
MVI A, 0FFH
STA 4526H
STA 4527H
HLT
Machine Code
LED Array
Metrics
Machine Code
Line Address Machine Code Source Code
1 0x0 3A 20 45 LDA 4520H
2 0x3 47 MOV B, A
3 0x4 3A 21 45 LDA 4521H
4 0x7 4F MOV C, A
5 0x8 21 00 00 LXI H, 0000H
6 0xB 79 MOV A, C
7 0xC B7 ORA A
8 0xD CA 15 00 JZ MULSTORE
9 MULLOOP:
10 0x10 09 DAD B
11 0x11 0D DCR C
12 0x12 C2 10 00 JNZ MULLOOP
13 MULSTORE:
14 0x15 22 22 45 SHLD 4522H
15 0x18 3A 24 45 LDA 4524H
16 0x1B B7 ORA A
17 0x1C CA 3C 00 JZ DIVZERO
18 0x1F 47 MOV B, A
19 0x20 3A 25 45 LDA 4525H
20 0x23 B7 ORA A
21 0x24 CA 3C 00 JZ DIVZERO
22 0x27 57 MOV D, A
23 0x28 78 MOV A, B
24 0x29 0E 00 MVI C, 00H
25 DIVLOOP:


Registers
A/PSW 0x FF 44
BC 0x 00 00
DE 0x 00 00
HL 0x 00 00
SP 0x 47 FF
PC 0x 00 59
Flags
S
Z
AC
P
C
V
K
Interrupts
TRAP
RST 5.5
RST 6.5
RST 7.5
; 16-bit Multiplication and Division
; Start Address: 4300H
LXI SP, 47FFH
; 16-bit multiplication
LHLD 4530H
XCHG
LHLD 4532H
MOV B, H
MOV C, L
LXI H, 0000H
SHLD 4536H
MOV A, B
ORA C
JZ MUL16STORE
MUL16LOOP:
DAD D
JNC NO_CARRY
PUSH H
LHLD 4536H
INX H
SHLD 4536H
POP H
NO_CARRY:
DCX B
MOV A, B
ORA C
JNZ MUL16LOOP
MUL16STORE:
SHLD 4534H
; 16-bit division
LHLD 453AH
MOV A, H
ORA L
JZ ERROR16
Machine Code
LED Array
Metrics
Machine Code
Line Address Machine Code Source Code
1 ; 16-bit Multiplication and Division
2 ; Start Address: 4300H
3
4 0x0 31 FF 47 LXI SP, 47FFH
5
6 ; 16-bit multiplication
7 0x3 2A 30 45 LHLD 4530H
8 0x6 EB XCHG
9 0x7 2A 32 45 LHLD 4532H
10 0xA 44 MOV B, H
11 0xB 4D MOV C, L
12 0xC 21 00 00 LXI H, 0000H
13 0xF 22 36 45 SHLD 4536H
14
15 0x12 78 MOV A, B
16 0x13 B1 ORA C
17 0x14 CA 2A 00 JZ MUL16STORE
18
19 MUL16LOOP:
20 0x17 19 DAD D
21 0x18 D2 24 00 JNC NO_CARRY
22 0x1B E5 PUSH H
23 0x1C 2A 36 45 LHLD 4536H
24 0x1F 23 INX H
25 0x20 22 36 45 SHLD 4536H